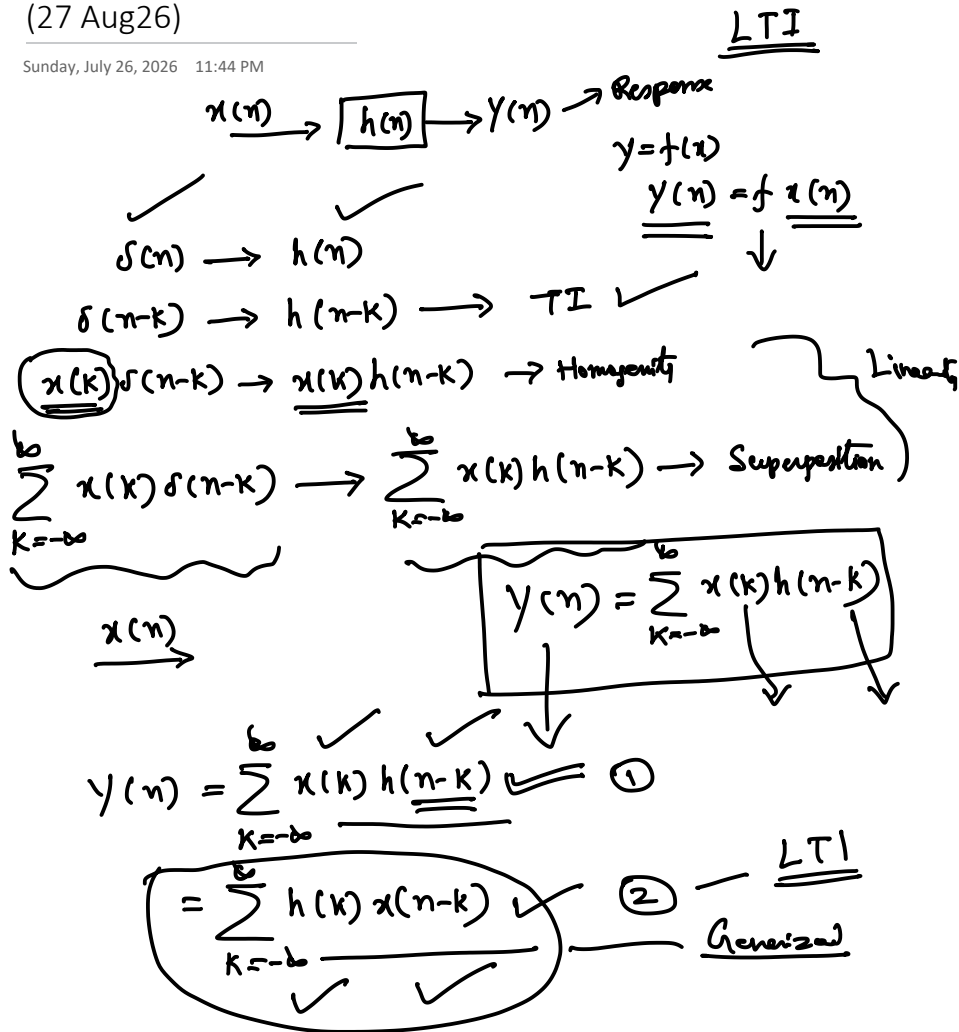


(27 Aug 26)

Sunday, July 26, 2026 11:44 PM

Note

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n,k)$$

$\rightarrow \underline{h(n-k)}$

$$\underline{h(n,k) \neq h(n-k)}$$

Time Varying System

Special CaseLTI C

1) DS is causal $\Rightarrow h(n) = 0, n < 0$

$$y(n) = \sum_{k=-\infty}^n x(k)h(n-k)$$

$\rightarrow \underline{k > n} \quad h(n-k) = \underline{\text{Arg}} = -ve = 0$

2) if $x(n)$ is causal

$y(n) = \sum_{k=0}^n x(k)h(n-k)$

LTI C (SES)

order

$$x(n) = \{ \underset{\uparrow}{3}, 2, 1, 0, 5, 2, \dots \}$$

Compute Conv. operation

$$x(k), h(k)$$

$$y(n) = \sum_{k=-\infty}^{\infty} x(k) h(n-k)$$

- 1) Fold any one sequence
- 2) Shift the folded sequence (delay Advance)
- 3) Point to point multiplication
- 4) Adding all products

$$x \quad \star$$

$$1) \text{ Commutative } \Rightarrow x(n) \star h(n) = h(n) \star x(n)$$

$$2) \text{ Associative } \Rightarrow x_1(n) \star x_2(n) \star x_3(n) = [x_1(n) \star x_2(n)] \star x_3(n) \\ = x_1(n) \star [x_2(n) \star x_3(n)]$$

$$3) \text{ Distributive } = x_1(n) \star [x_2(n) \star x_3(n)]$$

$$= x_1(n) \star x_2(n) + x_1(n) \star x_3(n)$$

$$\begin{array}{cccccc} & & x & & & \\ & & x(k), h(k) & & & \\ \hline -2 & -1 & 0 & 1 & 2 & \\ x_{-2} & x_{-1} & x_0 & x_1 & x_2 & - x(k) \checkmark \\ \hline h_{-2} & h_{-1} & h_0 & h_1 & h_2 & - h(k) \end{array}$$

$$y(n) = \sum_{k=-\infty}^{\infty} x(k) h(n-k) \\ n=0$$

$$y(n) = y(0) = x_0 h_0 + x_1 h_{-1} + x_2 h_{-2} \\ + x_{-1} h_1 + x_{-2} h_2$$

$$h_2 \quad h_1 \quad h_0 \quad h_{-1} \quad h_{-2} \quad - h(-k) \checkmark$$

$$y(1) = x_{-1} h_2 + x_0 h_1 + x_1 h_0 + x_2 h_{-1}$$

$$\underline{h_3} \quad \underline{h_2} \quad \underline{h_1} \quad \underline{h_0} \quad \underline{h_{-1}} \quad - h(1-k) \checkmark$$

$$y(-1) = x_{-2} h_1 + x_{-1} h_0 + x_0 h_{-1} + x_1 h_{-2}$$

$$\underline{h_1} \quad \underline{h_0} \quad \underline{h_{-1}} \quad \underline{h_{-2}} \quad \underline{h_{-3}} \quad - h(-1-k)$$

$$\begin{array}{l} N_1 = \text{len}(x) \\ N_2 = \text{len}(h) \\ \text{len}(y) \left\{ \begin{array}{l} N = N_1 + N_2 - 1 \end{array} \right. \end{array}$$

Easy Computation of Conv.

$$\begin{array}{rcccl} x(k) \rightarrow & & x_{-1} & x_0 & x_1 & x_2 \\ h(k) \rightarrow & \underline{h_{-2}} & \underline{h_{-1}} & \underline{h_0} & \underline{h_1} & \\ \hline & (h_{-1} x_{-1}) & h_{-1} x_0 & h_{-1} x_1 & h_{-1} x_2 & \end{array}$$

$$\begin{array}{ccccccc}
 & & h_0 x_{-1} & \left| \begin{array}{c} \underline{h_0 x_0} \\ h_{-1} x_1 \\ h_{-2} x_2 \end{array} \right| & h_0 x_1 & h_0 x_2 & x \\
 & h_{-1} x_{-1} & h_{-1} x_0 & & h_{-1} x_1 & h_{-1} x_2 & x \\
 h_{-2} x_{-1} & h_{-2} x_0 & h_{-2} x_1 & & h_{-2} x_2 & & x
 \end{array}$$

$$\begin{array}{ccccccc}
 \underline{y_{-3}} & y_{-2} & y_{-1} & y_0 & \underline{y_1} & y_2 & y_3
 \end{array}$$

— X —

$x(k)$
 $h(k)$

	x_{-1}	x_0	x_1	x_2		y_{-3}	y_{-2}	y_{-1}	y_0
h_{-2}	$x_{-1}h_{-2}$	x_0h_{-2}	x_1h_{-2}	x_2h_{-2}					
h_{-1}	$h_{-1}x_{-1}$	$h_{-1}x_0$	$h_{-1}x_1$	$h_{-1}x_2$					
h_0	h_0x_{-1}	<u>h_0x_0</u>	h_0x_1	h_0x_2					
h_1	h_1x_{-1}	h_1x_0	h_1x_1	h_1x_2					

— X —

Interconnection of systems

